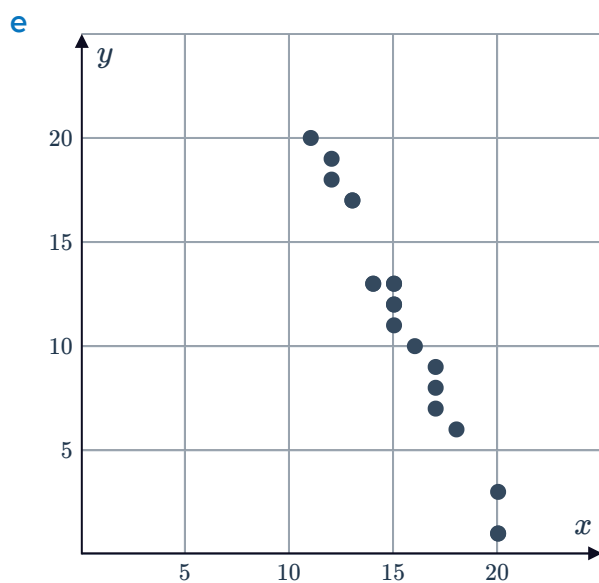
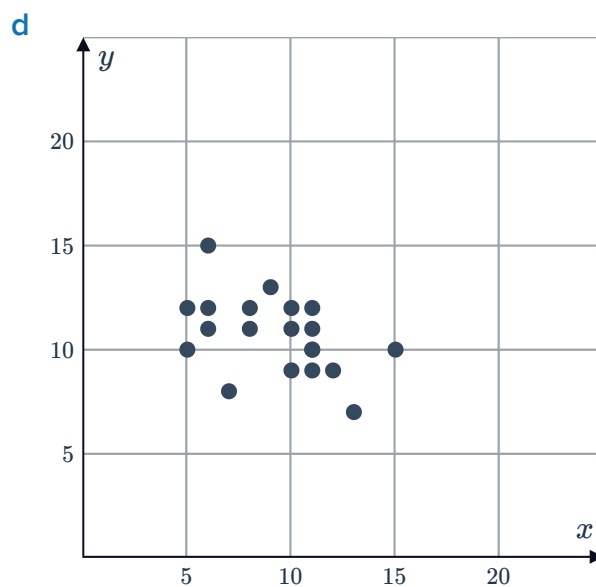
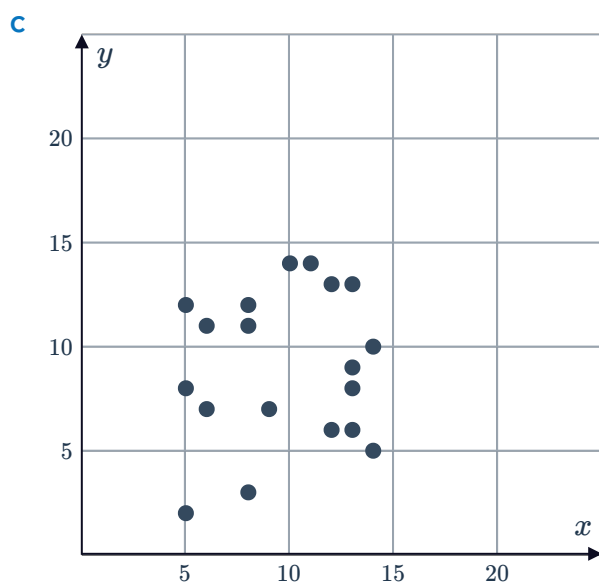
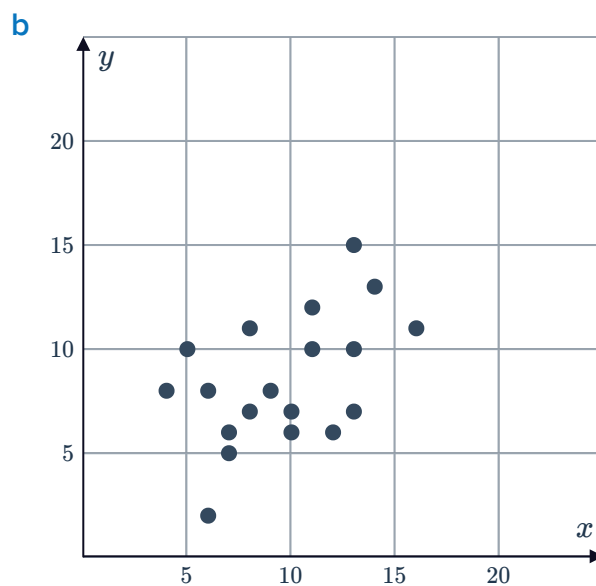
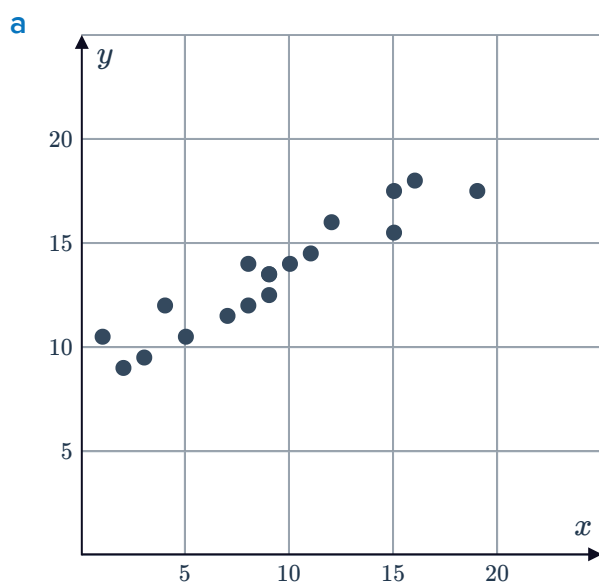


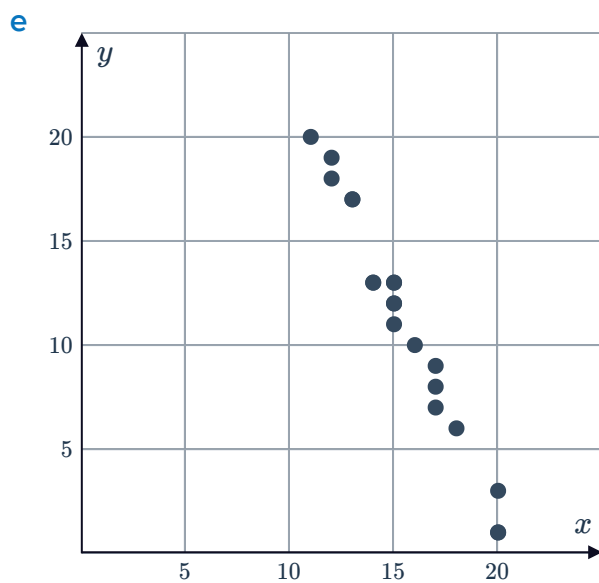
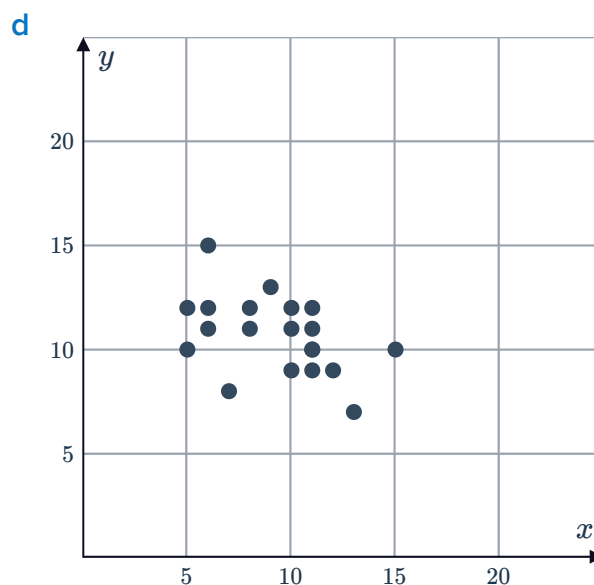
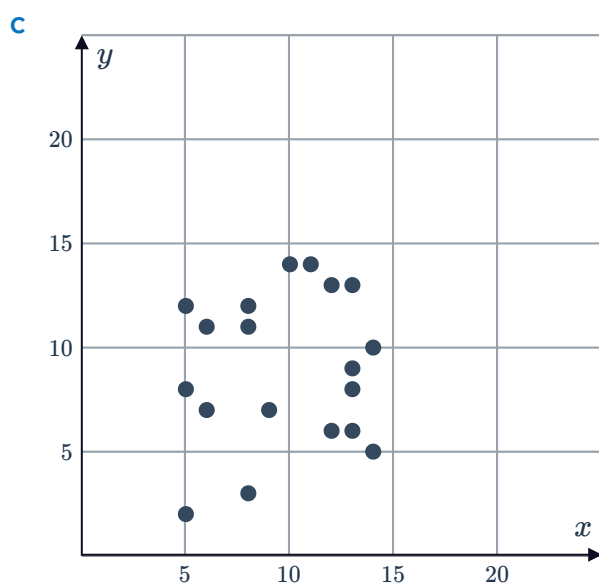
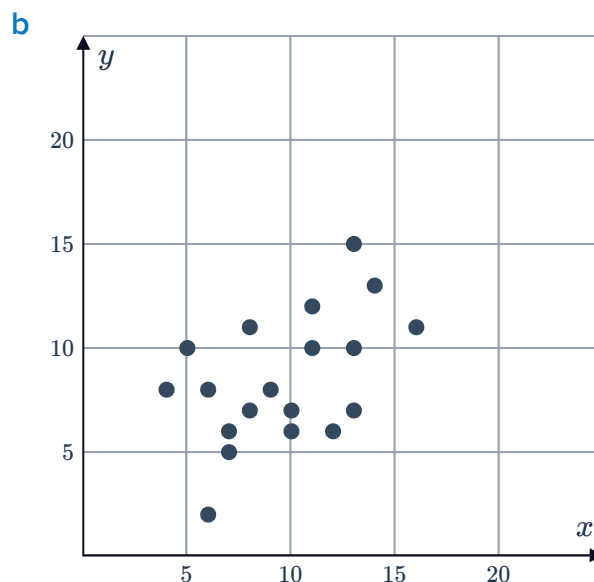
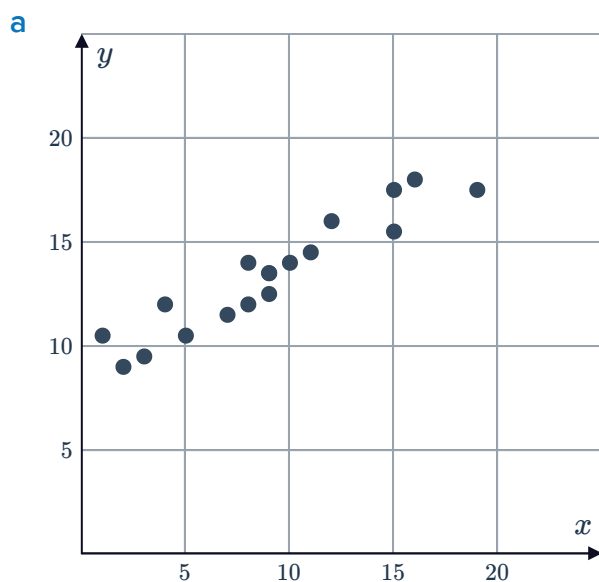
Pearson's correlation coefficient



1 Describe the type of linear relationship between the variables in the following scatter plots:



2 Estimate the value of the correlation coefficient for the data in the following scatter plots:



3 Describe the linear relationship between two variables with the following correlation coefficients:



- a 0.96 b 0.66 c 0.36 d -0.06
e -0.34 f -0.66

4 If the explanatory variable increases, describe the effect on the response variable for the following studies:

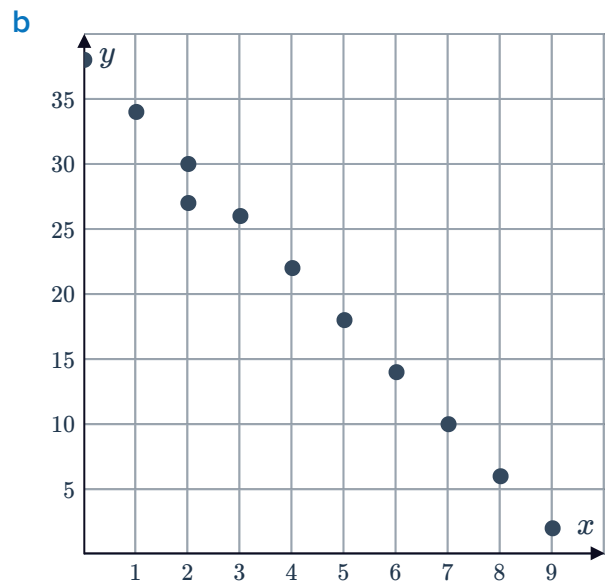
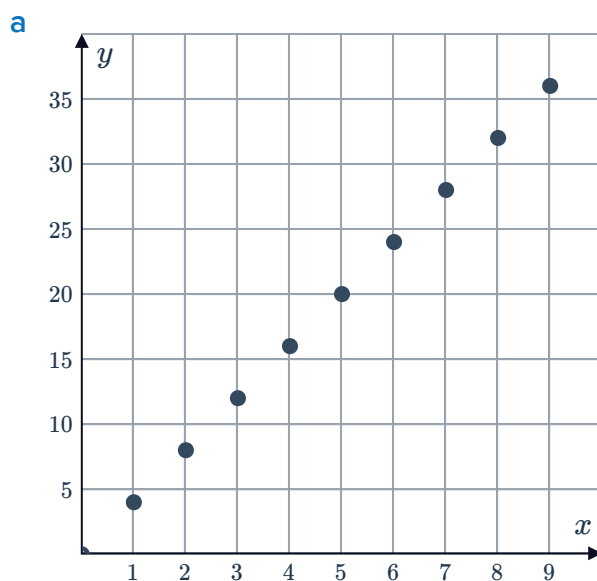
- a A study found that the correlation coefficient between heights of women and probability of being turned down for a promotion was found to be -0.90 .
b A study found that the correlation coefficient between population of a city and number of speeding fines recorded was found to be 0.83 .
c A study found that the correlation coefficient between length of hair and length of fingernails was found to be 0.07 .
d A study found that the correlation coefficient between number of bylaws a council has about dog breeding and number of dogs available for adoption at the local shelter was found to be 0.55 .

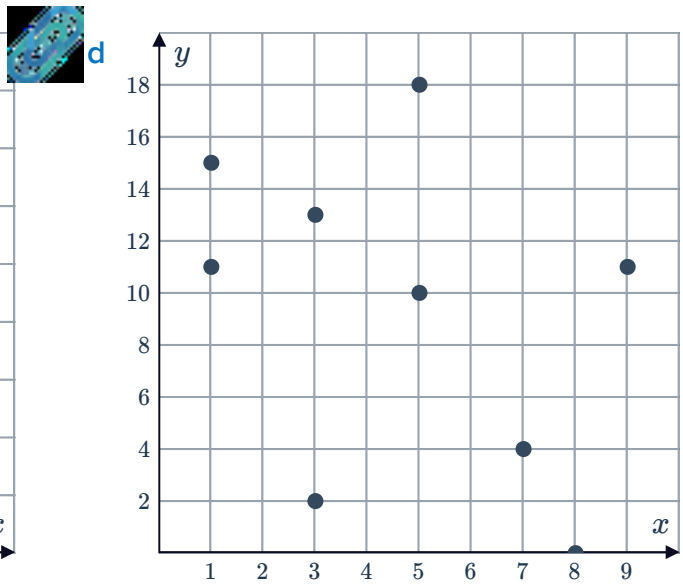
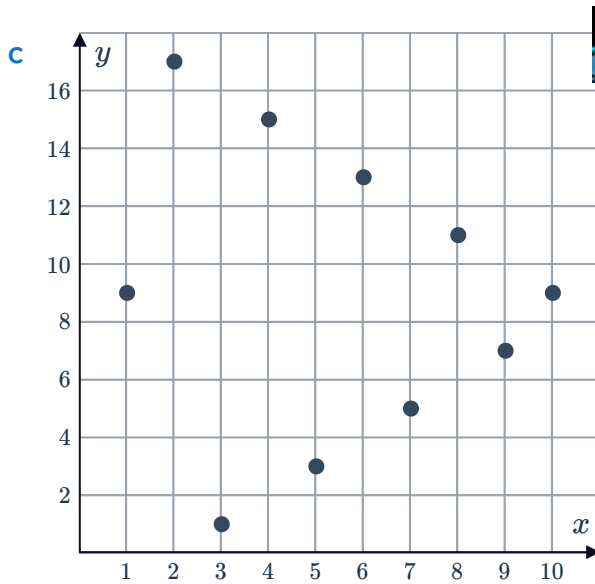
5 A pair of data sets have a correlation coefficient of $\frac{1}{10}$ while a second pair of data sets has a correlation coefficient of $\frac{3}{5}$. Which pair of data sets have the stronger correlation?

6 Describe the type of correlation the following correlation coefficients indicate:

- a $r = 1$ b $r = 0$ c $r = -1$

7 For each of the following graphs, write down an estimate of the correlation coefficient:





8 For each of the following pairs of relationships:

- i** Determine whether the two linear relationships have the same direction.
 - ii** State the relationship that has the stronger correlation.
- a** The linear relationship between a set of data for variables x and y has a correlation coefficient of 0.3. The linear relationship between a set of data for variables x and z has a correlation coefficient of 0.9.
 - b** The linear relationship between a set of data for variables x and y has a correlation coefficient of -0.9 . The linear relationship between a set of data for variables y and z has a correlation coefficient of -0.5 .
 - c** The linear relationship between a set of data for variables x and y has a correlation coefficient of 0.8. The linear relationship between a set of data for variables s and t has a correlation coefficient of -0.5 .
 - d** The linear relationship between a set of data for variables x and y has a correlation coefficient of 0.3. The linear relationship between a set of data for variables x and t has a correlation coefficient of -0.9 .

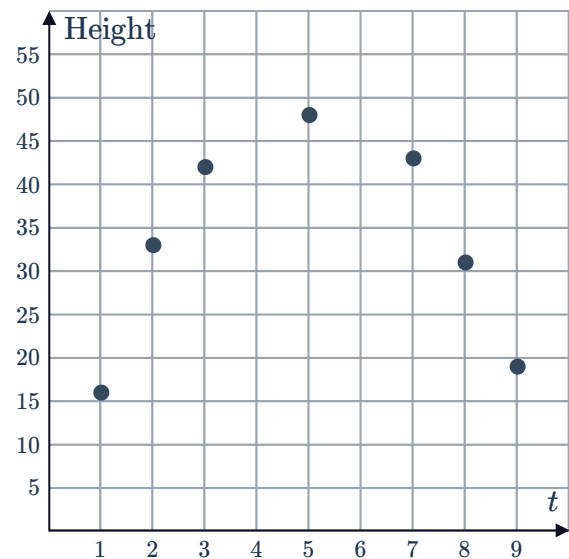
9 A researcher plotted the life expectancy of a group of men against the number of cigarettes they smoke a day. The results were recorded and the correlation coefficient r was found to be -0.88 .

Describe the correlation between the life expectancy of a man and the number of cigarettes smoked per day.

- 10 A researcher was evaluating the relationship between the number of years in education a person completes and the number of pets they own. The results were recorded and correlation coefficient r was found to be -0.3 . Describe the correlation between a person's years of education and the number of pets they own.

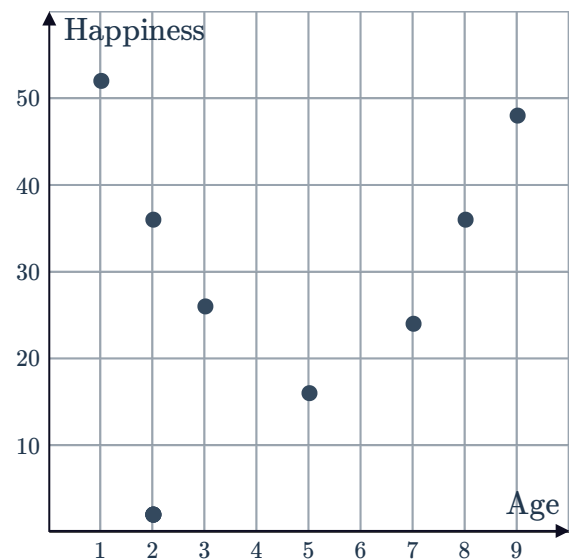
- 11 The scatter diagram shows data of the height of a ball kicked into the air as a function of time:

- a Which type of model is appropriate for the data, linear or non-linear?
- b Write down a possible value of Pearson's correlation coefficient, r , for this set of data.

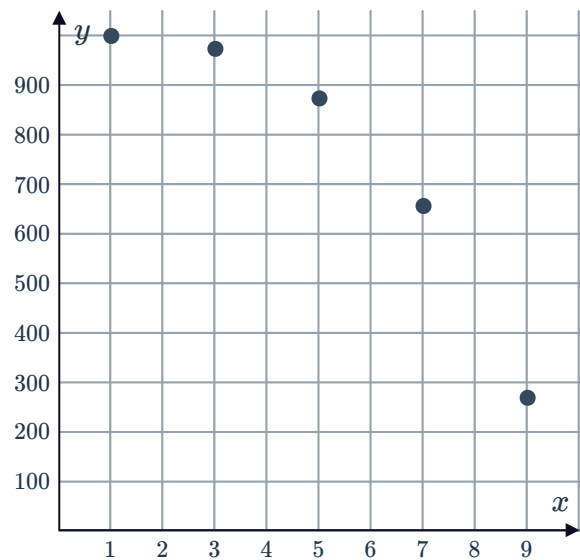


- 12 The scatter diagram shows data of a person's level of happiness as a function of their age:

- a Which type of model is appropriate for the data, linear or non-linear?
- b Write down a possible value of Pearson's correlation coefficient, r , for this set of data.



- 13** The scatter diagram shows data of the height of an object after it is pushed off a rooftop as a function of time:



- Which type of model is appropriate for the data, linear or quadratic?
- Write down a possible value of Pearson's correlation coefficient, r , for this set of data.

Correlation vs causation

- 14** Determine whether the following describe a relationship that is correlated but not causal:
- The sales of ice cream and increase in temperature.
 - The number of hours worked and how much money is made for a given person.
 - The amount of showers had in a day and the amount of the water bill.
 - The amount of rainfall received, and level of water in a lake.
 - The larger the dimensions of a rectangular verandah, the more area.
 - The season of the year and the number of water related injuries.
 - Increase in temperature, and the level of mercury in a thermometer.
 - The number of students shouting in class and the number of detentions received.
- 15** For each of the following data examples, determine if there is a causal relationship between the variables:
- The number of times a coin lands on heads and the likelihood that it lands on heads on the next flip.
 - The amount of weight training a person does and their strength.
- 16** Determine whether the following describe a causal relationship and not just a correlation:
- An individual's decision to work in construction and his diagnosis of skin cancer.
 - The number of minutes spent exercising and the amount of calories burned.
 - A decrease in temperature and the increase in attendance at an ice skating rink.
 - As a child's weight increases, so does her vocabulary.

17 Determine whether the following are examples of variables with no correlation:



- a The age of a child and their shoe size.
- b The age of a child and their height.
- c The age of a child and the number of pets owned.
- d The age of a child and the amount of adjectives learned.

18 The table shows the number of fans sold at a store during days of various temperatures:

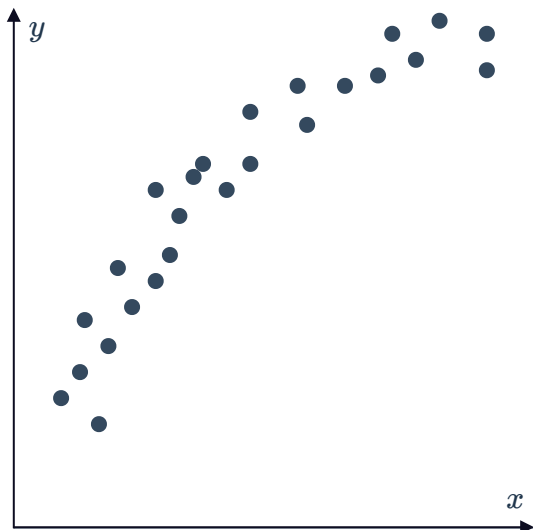
Temperature (°C)	6	8	10	12	14	16	18	20
Number of fans sold	12	13	14	17	18	19	21	23

- a For this data, will r be greater than zero, less than zero or equal to zero?
 - b Is there a causal relationship between the variables? Explain your answer.
- 19 A study found a strong positive association between the temperature and the number of beach drownings.
- a Does this mean that the temperature causes people to drown? Explain your answer.
 - b Is the strong correlation found a coincidence? Explain your answer.
- 20 Many trees lose their leaves in winter. Does this mean that cold temperatures cause the leaves to fall?
- 21 A study found a strong correlation between the approximate number of pirates out at sea and the average world temperature.
- a Does this mean that the number of pirates out at sea has an impact on world temperature?
 - b Is the strong correlation found a coincidence? Explain your answer.
 - c If there is correlation between two variables, is there causation?

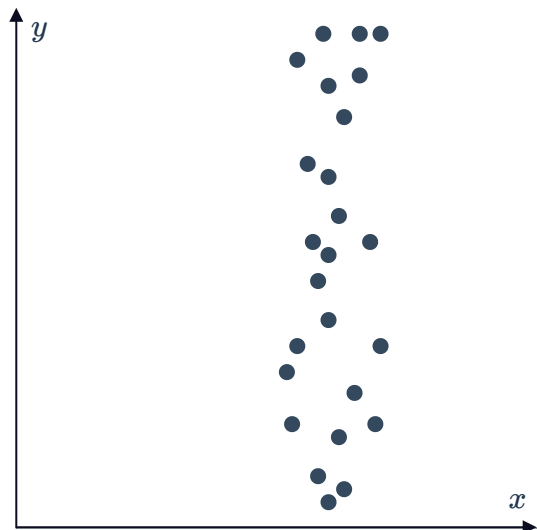
22 Scatter plots for two sets of data are shown below:



Set A



Set B



- a Which data set has the strongest linear correlation between the variables?
- b Which data set appears to have a non-linear relationship? Explain your answer.
- c Why does data set B have the weakest correlation between its variables?
- d State whether the following pairs of variables could be represented by data set B:
 - i Marks in an English examination and distance travelled from home to school.
 - ii Cost of cars and cost of petrol.
 - iii Distance travelled in a car and the cost of a driver's license.
 - iv Height and weight of students at school.